Chen 2020 NPC — analysis summary

RUN 4 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/14 samples pass the min-cell threshold 107; Harmony(sample, θ=2) fully intersperses all 8 patients; 1 connected component after island removal (438 detached cells dropped). Mixed and continuous.

Gate 2 — module geography → NO-GO

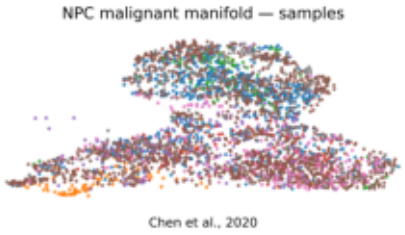
A robust Crest-terminal axis exists (Crest opposes Gast −0.64 / pSom −0.44, SURVIVES cell-cycle correction → not a proliferation artifact). But the defining embryonic BODY is absent: Gast/pSom are decoupled (r=+0.01 vs Kürten GO +0.33 / Puram GO −0.49) and pSom does not participate (r(t,pSom)=−0.17, flat). A two-module Gast/Crest gradient without a coupled Gast/pSom body is not the embryonic pattern.

Gate 3 — hub cross-repression → NA

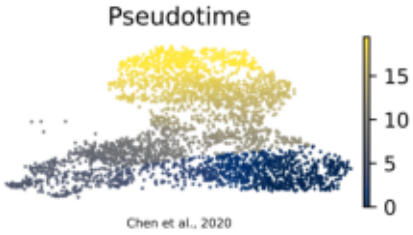
No bifurcation on a linear manifold → no Gast↔pSom arms to test. Correlation + modules notebooks not produced.

STEP FIGURES

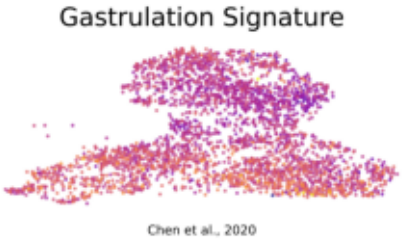
Manifold — samples



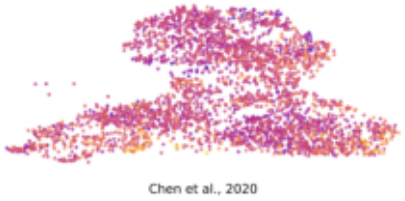
Pseudotime



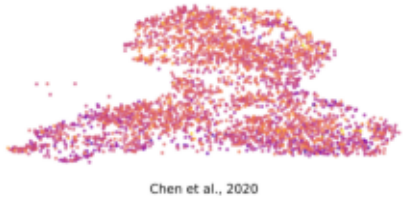
Gastrulation



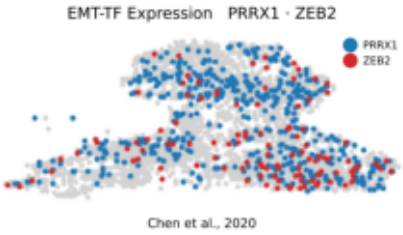
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 3,236 malignant cells (8 samples).

Verdict.
NOT REPRODUCED (embryonic geography absent; Gate 2 NO-GO). Malignant NPC cells form a mixed, continuous, non-branching manifold with a real, cell-cycle-independent Crest-terminal gradient (Crest opposes Gast −0.64 / pSom −0.44 after cc-correction). But the three-module embryonic pattern is not present: Gast/pSom decoupled (r=+0.01, vs Kürten GO +0.33) and pSom does not participate (flat, r(t,pSom)=−0.17). This NO-GO corrects, rather than echoes, the prior attempt's 'generic-carcinoma' rationale — the axis is embryonic-flavoured and NOT mere proliferation; what fails is the coupled Gast/pSom body the gate requires.

Parameter choices

Expression	UMI → normalize_total, log1p (no scale)	**Threshold**	min 107 malignant cells/sample (gap 36→107, ×2.97) → 8 samples, 3,674 cells
Embedding	overdispersed → PCA(19 PCs, >0.5% var) → Harmony(sample, θ=2) → knn15 → UMAP → Leiden 0.3	**Island removal**	438 cells (clusters 2,4,6,7,8) → 3,236 cells, 1 component
Subtypes	Cycling, Protein-maturation, Hypoxia; 3CA meta-programs + Chen calls	**Manifold**	scFates curve on X_umap (non-branching; 0 branch nodes, 12-combo PPT sweep); rooted Gast/pSom-high
EMT-TFs shown	SOX4/PRRX1/TCF4/SNAI2/ZEB2/FOXC1 + 6 more (detect≥10%); canonical SNAI1/ZEB1/TWIST1 dropped		